



2.11

Logarithmic Functions

Student Notes and Guided Practice

AP Precalculus - Unit 2

| ECHS Mathematics

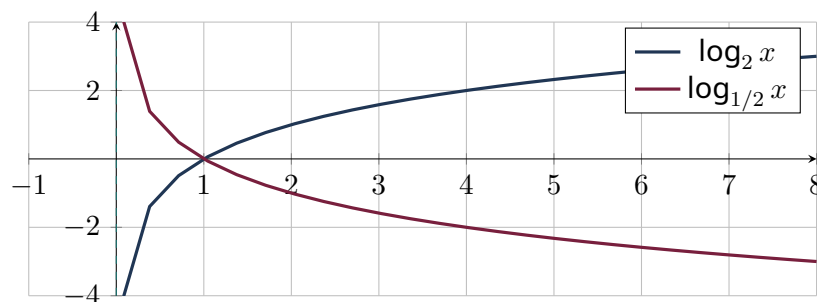
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Learning Targets

- Analyze logarithmic domain, range, asymptote, monotonicity, and concavity.
- Interpret equal multiplicative input changes as equal additive output changes.
- Analyze transformed logarithmic functions.

Essential Question

How does a logarithmic function convert multiplicative input structure into additive output structure?



Concept Framework

Key Idea

For base greater than 1, log is increasing and concave down.

Key Idea

For base between 0 and 1, log is decreasing and concave up.

Key Idea

The parent domain is $(0, \infty)$, range all real numbers, vertical asymptote $x = 0$.

Key Idea

Transformations shift the vertical asymptote and change the domain.

Formula Map**Formula and Structure**

$$f(x) = a \log_b(x - h) + k$$

Formula and Structure

vertical asymptote $x = h$

Formula and Structure

$$\log_b(cx) - \log_b x = \log_b c$$

Worked Examples**Worked Example 1**

Problem. Analyze $f(x) = 2 \log_3(x + 4) - 1$.

Solution. Domain $(-4, \infty)$; range all real; vertical asymptote $x = -4$; increasing and concave down.

Worked Example 2

Problem. If x is multiplied by 9, how much does $\log_3 x$ change?

Solution. It increases by $\log_3 9 = 2$.

Worked Example 3

Problem. Why can a logarithmic function have any real output?

Solution. For every real y , $b^y > 0$, so some positive input has log value y .

Multiple-Representation Connection**AP Reasoning**

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Analyze $\log_2 x$: domain, range, asymptote, monotonicity, concavity.

2. Analyze $\log_{1/3} x$: domain, range, asymptote, monotonicity, concavity.

3. Analyze $2 \ln(x - 4) + 1$: domain, range, asymptote, monotonicity, concavity.

4. Analyze $-\log_5(x + 2) + 3$: domain, range, asymptote, monotonicity, concavity.

5. Analyze $\ln(7 - x)$: domain, range, asymptote, monotonicity, concavity.

6. Find x-intercept of $3 \log_2(x - 1) - 6$.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.